



AQ1.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

2.1 Level 1:

1 point

Suppose $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 11 & -29 & -6 & \\ -3 & 4 & 7 & 7 \end{bmatrix}$. Which of the following is true about the matrix A

A?

[Hint: The (i, j) -th entry is the entry which is at the i -th row and j -th column.]

☐

It is a $4 \times 3 \times 4$ matrix.

☒

It is a $3 \times 4 \times 3$ matrix.

☐

$(2, 3)$ -th entry of the matrix AA is 44.

☒

$(2, 3)$ -th entry of the matrix AA is 99.

Yes, the answer is correct.

Score: 1

Accepted Answers:

It is a $3 \times 4 \times 3$ matrix.

$(2, 3)$ -th entry of the matrix AA is 99.

1 point

Which of the following statements is(are) TRUE?

[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

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Any diagonal matrix is a scalar matrix.

☐

Scalar matrices may not be square matrices.

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Scalar matrices must be square matrices.

☐

Any scalar matrix is an identity matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Scalar matrices must be square matrices.

1 point

Given below is the system of linear equations:

$$7x_1 + 10x_2 + 12x_3 = 368x_1 + 4x_2 - 9x_3 = 114x_1 - x_2 + 3x_3 = 107x_1 + 10x_2 + 12x_3 \\ = 368x_1 + 4x_2 - 9x_3 = 114x_1 - x_2 + 3x_3 = 10$$

If the matrix representation of the system of equations is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

☒


$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 \\ 8 & 4 & -9 & & \\ 4 & -1 & 3 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 1 & 2 & -9 & 3 & & \\ & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & \\ 3 & 6 & 1 & 1 & 1 & 0 \\ & \end{bmatrix}$$

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$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 & 3 & 6 \\ 8 & 4 & -9 & 1 & 1 & 1 & \\ 4 & -1 & 3 & 1 & 0 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & 1 & 1 & 2 & 9 & 3 & 3 & 6 & 1 & 1 & 1 & 0 \\ & & & & & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & \\ 3 & 6 & 1 & 1 & 1 & 0 \\ & \end{bmatrix}$$

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$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 & 3 & 6 \\ 8 & 4 & -9 & 1 & 1 & 1 & \\ 4 & -1 & 3 & 1 & 0 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 1 & 2 & -9 & 3 & 3 & 6 & 1 & 1 & 1 & 0 \\ & & & & & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & \\ 3 & 6 & 1 & 1 & 1 & 0 \\ & \end{bmatrix}$$

○

$$A = \begin{bmatrix} 7 & 1 & 0 & 3 & 6 & 1 & 2 \\ 8 & 4 & -9 & 1 & 1 & 1 & \\ 4 & -1 & 1 & 0 & 3 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 3 & 6 & 1 & 1 & 0 & 1 & 2 & -9 & 3 & & \\ & & & & & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & \\ 3 & 6 & 1 & 1 & 1 & 0 \\ & \end{bmatrix}$$

Yes, the answer is correct.**Score: 1****Accepted Answers:**

$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 \\ 8 & 4 & -9 & & \\ 4 & -1 & 3 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 1 & 2 & -9 & 3 & & \\ & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & \\ 3 & 6 & 1 & 1 & 1 & 0 \\ & \end{bmatrix}$$

1 point

Which of the following statements is(are) TRUE?

☐ Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.

☒ Addition of two matrices is possible only if the orders of both the matrices are the same.

☐

Defining $ABAB$ is possible if the number of rows in the matrix AA is same as the number of columns in the matrix BB .

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Defining $ABAB$ is possible if the number of columns in the matrix AA is same as the number of rows in the matrix BB .

Yes, the answer is correct.**Score: 1****Accepted Answers:**

Addition of two matrices is possible only if the orders of both the matrices are the same.

Defining $ABAB$ is possible if the number of columns in the matrix AA is same as the number of rows in the matrix BB .

1 point

Suppose $P = \begin{bmatrix} 3 & -17 \\ 4 & 0 \\ 2 & -52 \end{bmatrix}$, $Q = \begin{bmatrix} 14 & -9 \end{bmatrix}$, $R = \begin{bmatrix} 0 & -310 \end{bmatrix}$, $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$

$D = \begin{bmatrix} \\ \\ -245 \end{bmatrix}$

[Hint: If AA is a matrix of order $m \times nm \times n$ and BB is a matrix of order $n \times pn \times p$, then the order of $ABAB$ is $m \times pm \times p$.]



The matrix $PDPD$ is of order $3 \times 1.3 \times 1$.



The matrix $PDPD$ is of order $1 \times 3.1 \times 3$.



The matrix $QDQD$ is of order $3 \times 3.3 \times 3$.



The matrix $QDQD$ is of order $1 \times 1.1 \times 1$.



The matrix $DQDQ$ is of order $3 \times 3.3 \times 3$.



The matrix $DQDQ$ is of order $1 \times 1.1 \times 1$.



The product $QDQD$ is not defined.



The product $QRQR$ is not defined.



The addition $P + QP + Q$ is not defined.



The addition $P + DP + D$ is not defined.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix $PDPD$ is of order $3 \times 1.3 \times 1$.

The matrix $QDQD$ is of order $1 \times 1.1 \times 1$.

The matrix $DQDQ$ is of order $3 \times 3.3 \times 3$.

The product $QRQR$ is not defined.

The addition $P + QP + Q$ is not defined.

The addition $P + DP + D$ is not defined.

2.2 Level 2:

1 point

Choose the set of correct options.

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given conditions to find matrix AA , for

first two options.]



If AA is a square matrix of order 2 and $A^2 = IA^2 = I$, then $A = IA = I$ or $A = -IA = -I$ where II is the identity matrix of order 2.



If AA is a square matrix of order 2 and $A^2 = 0A^2 = 0$, then $A = 0A = 0$.



If AA and BB are square matrices of order 3 and $A + B = 0A + B = 0$, then $B = -AB = -A$.



If AA is a scalar matrix of order 3, BB is a non-zero square matrix of order 3 and $AB = 0AB = 0$, then $A = 0A = 0$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If AA and BB are square matrices of order 3 and $A + B = 0A + B = 0$, then $B = -AB = -A$.

If AA is a scalar matrix of order 3, BB is a non-zero square matrix of order 3 and $AB = 0AB = 0$, then $A = 0A = 0$.

1 point

If AA is a square matrix of order 22 whose first column is denoted by C_1 and second column is denoted by C_2 (the left most column is taken as the first one) and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ then choose the set of correct options.

[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]



The first column of $ABAB$ is $b_{11}C_1 + b_{21}C_2$.



The second column of $ABAB$ is $b_{12}C_1 + b_{22}C_2$.



The first column of $ABAB$ is $b_{21}C_1 + b_{11}C_2$.



The second column of $ABAB$ is $b_{22}C_1 + b_{21}C_2$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The first column of $ABAB$ is $b_{11}C_1 + b_{21}C_2$.

The second column of $ABAB$ is $b_{12}C_1 + b_{22}C_2$.

1 point

Choose the set of correct options.

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given condition to find matrix AA for last two options.]



There exist some real matrices AA and BB , such that $AB = BAAB = BA$.



There do not exist any real matrices AA and BB , such that $AB = BAAB = BA$.



There does not exist any real matrix AA , such that $A^2 = AA^2 = A$.



There exists some real $2 \times 2 \times 2$ matrix AA , such that $A^2 + A + I = 0A^2 + A + I = 0$

Yes, the answer is correct.

Score: 1

Accepted Answers:

There exist some real matrices AA and BB , such that $AB = BAAB = BA$.

There exists some real $2 \times 2 \times 2$ matrix AA , such that $A^2 + A + I = 0A^2 + A + I = 0$

Suppose AA is a 3×3 scalar matrix and (1,1)-th entry of the matrix AA is 4. Suppose BB is a 3×3 square matrix such that (i, j) -th entry is equal to $i^2 + j^2$. Find the (2,2)-th entry of the

matrix $2A + B$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 16

1 point

Suppose $A = \begin{bmatrix} -1 & 2 \\ 4 & 7 \end{bmatrix}$ and $A^2 - \alpha A + I = 0$ for some $\alpha \in \mathbb{R}$.

Find the value of α .

Yes, the answer is correct.

Score: 1

Accepted Answers: